

On the QUD sensitivity of a third reading

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Introduction

- Experimental evidence for the hypothesis that availability of a transparent reading is sensitive to the QUD (Question Under Discussion) [1–5], focusing on different attitudinal QUDs [4, 5].
→ Consequence for (replacement) theories of transparent readings.
- Methodological contribution: How do we manipulate a QUD about someone’s mental state in experimental settings?

Factual QUDs and transparent readings

- Attitude reports can address *attitudinal* or *factual* QUDs [6–12]:
 - a. *Attitudinal* QUD: Why did Ana not ask Bob out?
Ana thinks that Bob is dating Cindy.
 - b. *Factual* QUD: Who is Bob dating?
Ana thinks that Bob is dating Cindy.
- Factual QUDs are known to facilitate transparent readings [1–3].
 - Percus [2] proposes a replacement mechanism that require factual QUDs for special cases of transparent readings. He asks which other cases it extends to.

Attitudinal QUDs and transparent readings

- Mayr & Schmitt [4, 5] claim that attitudinal *existential* QUDs also license transparent readings (2).
- (2) Scenario: Joe went to a party. He thought all guests were biologists. Except for three linguists, Ann, Bea, and Cate, this was the case. Joe doesn’t know them or their names. He saw each of them dancing with Eve, who he knows. So he thinks Eve and one of them are lovers but is unsure which of them. *Eve’s relationship status was discussed.*
 - a. ✓ Joe thinks that Eve is involved with a **linguist**.
 - b. Hypothesized QUD: Does Joe think that Eve is single?
- In their proposal [4, 5], replacement of an element in the attitude complement is allowed if the attitude report entails the same answer to the QUD before and after the replacement.
- (3) Joe thinks that Eve is involved with {a linguist, a person dancing with Eve} → No to (2b).

Experiment design & Task

- 2×2 : QUD (attitudinal: *existential* vs. *wh*) × transparency
- PCIbex survey, Recruitment via Prolific, 356 (Story A) / 359 (Story B) / 345 (Story C) participants after exclusion.
- ❶ Participants read a story (see “Stimuli”).
- ❷ A, who read only the beginning of the story, asks B, who read the whole story, a question about the rest of the story.
- ❸ Participants rate whether they “Agree” or “Disagree” with B’s response (= the target attitude report, **string-identical across conditions** but only true under the intended transparency).
- All factors tested between-subject, one story per participant. Why?
 - Individual items involve long stories: this is by necessity to set up a discourse that both supports the attitude statement and the attitudinal QUD.
 - Pilot research shows that QUD effects on transparency readings are subtle.
 - One item per participant means no carry-over effects in interpretation resolution.

Stimuli

(4) Story A, {**Existential QUD**, *Wh QUD*} × **Transparent reading**

Mia moved into a new apartment last week. {Her floor is very quiet, so she was wondering if the apartment next door is vacant. It is in fact vacant, but Mia didn’t know that for a fact. / She knew that the apartment next door is occupied, but she didn’t know anything about the person who lives there. She was hoping to learn more about her neighbor. In particular, Mia was wondering what her next-door neighbor does for work. Unbeknownst to Mia, her neighbor is actually a carpenter.}

Yesterday, before she left to go furniture shopping for her new apartment, Mia saw a group of three men in the hallway. {The men were each viewing the apartment next door, / They were actually visiting Mia’s neighbor,} and **they all happened to be programmers. However, Mia didn’t know any of that.** Now Mia thinks that one of the men she saw in the hallway yesterday lives in the apartment next door, although she isn’t sure which man.

Ann: I read that Mia was wondering {if the apartment next door is vacant / what her next-door neighbor does for work}. What does she think now?
Bob: She thinks that a **programmer** lives next door.

Recipes for QUD manipulation

- Knowledge gap between two discourse agents. (e.g., [13–16])
- Inquisitive attitude set up by a *wondering* sentence.
- The attitude holder gains a wrong belief (cf. [17, 18]).
- Factual QUDs and the QUD of the other condition are ruled out.

Results

- A Bayesian mixed-effects logistic regression model (rating~QUD*transparency+(1|story); reference: QUD=*wh*, Transparency = transparent) shows significant main effects of QUD ($\beta=1.23$, HPD=[0.83, 1.58]) and transparency ($\beta=2.35$, HPD=[1.88, 2.87]) as well as interaction ($\beta=-0.83$, HPD=[−1.59, −0.09]).

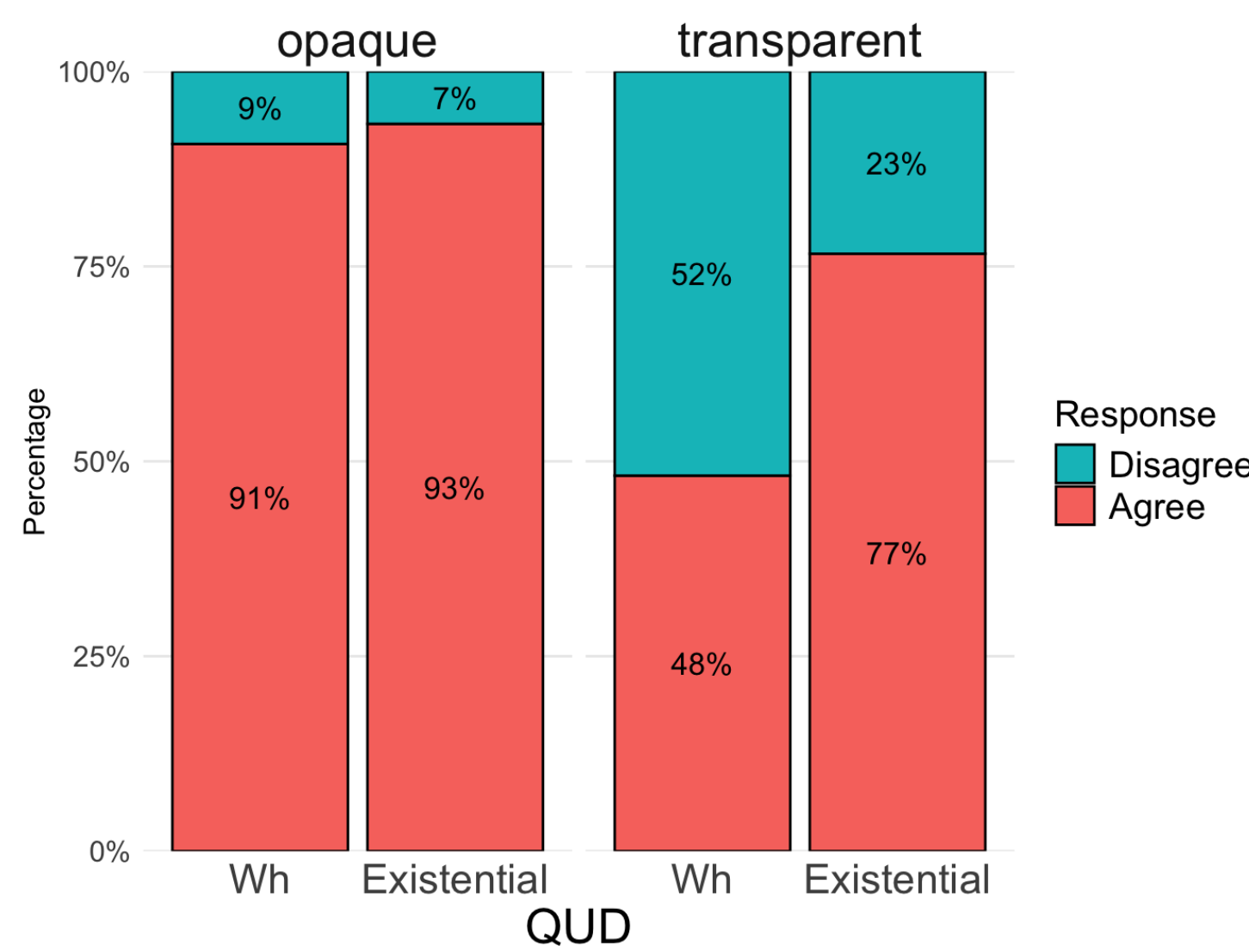


Figure 1: Response distribution by QUD and transparency across three stories

- Same, consistent pattern in each of the three stories.

Discussion

- We found support for Mayr & Schmitt’s empirical claim that transparent readings are available under some attitudinal QUDs [4, 5].
- Still, if there is no semantic distinction between transparent and opaque readings [4, 5], why do transparent readings tend to be less available?
 - Due to non-trivial replacement? (Viola Schmitt, p.c.)
- Zhang & Davidson [19] found no difference between *de re* and *de dicto* readings in analogous contexts without overt QUD manipulation.
 - What was/were the implicit QUD(s) in their studies?
- QUD sensitivity does not rule out non-replacement theories; it’s compatible with transparent evaluation theories [20–22].

- a. Existential QUD: Does M. think the apartment next door is vacant?
 $\approx \{\lambda w. \forall w' \in \text{DOX}_{M,w}. \exists x[x \text{ lives next door in } w'],$
 $\lambda w. \forall w' \in \text{DOX}_{M,w}. \neg \exists x[x \text{ lives next door in } w']\}$
- b. [She thinks that a programmer lives next door] $^{w,g} =$
 $\forall w' \in \text{DOX}_{M,w}. \exists x[[\text{programmer}]^{w,g}(x) \wedge x \text{ lives next door in } w']$

[1] M. Schwager. Speaking of qualities. SALT 19, 2009. [2] O. Percus. Index-Dependence and Embedding. 2020. [3] I. Benbaji-Elhadad. Preliminaries for a substitution of de re. SALT 33, 2023. [4] C. Mayr and V. Schmitt. Non-de dicto construals as a unified phenomenon. PhilP Workshop 8, 2023. [5] C. Mayr and V. Schmitt. Non-de dicto construals as a unified phenomenon. SALT 34, 2024. [6] J. O. Urmson. Parenthetical verbs. Mind, 1952. [7] J. B. Hooper. On assertive predicates. Syntax and Semantics, 1975. [8] M. Simons. Observations on embedding verbs, evidentiality, and presupposition. Lingua, 2007. [9] D. Beaver. Have you noticed that your belly button lint color is related to the color of your clothing? Presuppositions and discourse, 2010. [10] M. Simons, D. Beaver, C. Roberts, and J. Tonhauser. The best question: Explaining the projection behavior of factives. Discourse processes, 2017. [11] S. AnderBois. Semantics and pragmatics of (not-) at-issueness in Yucatec Maya attitude reports. Semantics and Pragmatics, 2016. [12] D. Beaver, C. Roberts, M. Simons, and J. Tonhauser. Questions under discussion: Where information structure meets projective content. Ann. Rev. of Linguistics, 2017. [13] A. Cremers, E. Coppock, J. Dotlacil, and F. Roelofsen. Modified numerals: Two routes to ignorance. 2017. [14] L. Kursat and J. Degen. Probability and processing speed of scalar inferences is context-dependent. CSS 42, 2020. [15] E. Ronai and M. Xiang. Pragmatic inferences are QUD-sensitive: An experimental study. J. of Linguistics, 57, 2021. [16] B. Grusdt, M. Liu, and Michael Franke. Testing the influence of QUDs on the occurrence of conditional perfection. Experiments in Linguistic Meaning, 2023. [17] S. Lewis, J. Lidz, and V. Hacquard. The semantics and pragmatics of belief reports in preschoolers. SALT 22, 2012. [18] V. Hacquard and J. Lidz. Children’s attitude problems: Bootstrapping verb meaning from syntax and pragmatics. Mind & Language, 2019. [19] Y. Zhang and K. Davidson. Interpreting referential noun phrases in belief reports—the de re/de dicto competition. Glossa Psycholinguistics, 2024. [20] K. von Fintel and I. Heim. Intensional semantics. MIT, 2011. [21] E. Keshet. Good intensions: Paving two roads to a theory of the de re/de dicto distinction. MIT, 2008. [22] Ö. Demirok. Scope theory revisited: lessons from pied-piping in wh-questions. MIT, 2019.